

Q4		
(a)(i)	(Use FLHR to get the direction of force – the force will provide for centripetal acceleration for circular motion as it is always perpendicular to the velocity and hence points towards centre of circle. The center of circle should be below point A) <u>Complete circle with centre of circle vertically below point A and the arrow tangential to the circle.</u>	B1
(a)(ii)	The <u>magnetic force provides for the centripetal force</u> required for circular motion,	B1
	$Bqv = \frac{mv^2}{R} \Rightarrow R = \frac{mv}{Bq} = \frac{(9.11 \times 10^{-31} \text{ kg})(2.8 \times 10^7 \text{ m s}^{-1})}{(1.2 \times 10^{-3} \text{ T})(1.60 \times 10^{-19} \text{ C})} = 0.133 \text{ m}$ = 13 cm (shown)	M1
(b)(i)1	Direction of electric field is downward in the plane of the paper.	A1
(b)(i) 2	For the electron to be undeflected, the net force on it must be zero. $qvB = qE$ $\Rightarrow E = vB = (2.8 \times 10^7)(1.2 \times 10^{-3})$ $= 3.36 \times 10^4 \text{ N C}^{-1}$	M1 A1
(b)(ii)	<u>Helix</u> (out of the plane of the paper) with an <u>increasing pitch</u> <i>Remarks :</i> - Spiral not accepted as answer. Spiral is not helix – a spiral has a changing radius. “Increasing pitch” can be marked from the diagram if it is included. But at least four turns needs to be drawn for any credit.	B1 B1

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